

⚡ FUN WITH ELECTRICITY! ⚡

Voltage Divider & Potentiometer: Manual Dimmer Fun!

💧 Imagine Electricity Flowing Like Water!

Have you ever turned a water faucet or tap? Turning the knob lets more water flow out, and turning it back slows it down to a trickle! **Electricity works in a very similar way!** Just like water needs pipes to flow through, electricity needs wires (paths) to move from one place to another to power our lights, fans, and gadgets.

💡 What is Electricity?

Electricity is a form of energy that powers our world! It travels through circuits (paths) to make things work:

- **Battery:** The power source that gives electricity its energy push!
- **Wires:** The pipes through which electricity flows.
- **Light Bulb / Fan:** The cool gadgets that use electricity to glow bright or spin!

🔌 What is a Voltage Divider & Potentiometer?

Component	What It Does
Voltage Divider	Splits up or divides electricity in a path so we can control how much energy goes to a light bulb or fan.
Potentiometer	A special Magic Knob! Turning it changes how easily electricity flows through the circuit.
Resistors in Series	Two or more resistors connected one after another to help control electrical flow step-by-step.

✨ How Does a Potentiometer Work as a Dimmer?

When you turn the potentiometer knob, you create the **Dimmer Effect**:

- **Turn Clockwise (More Turn):** Lets **MORE** electricity flow $\rightarrow$ Light gets **BRIGHTER!** ☀️
- **Turn Counter-Clockwise (Less Turn):** Lets **LESS** electricity flow $\rightarrow$ Light gets **DIMMER!** 🌙



Hands-On Experiment: Build Your Dimmer!

1. Connect the **battery**, **potentiometer**, and **light bulb** in a loop.
2. Turn the knob slowly and watch the light change from dim to bright!
3. Observe how turning the knob changes the brightness just like a ceiling fan regulator changes speed!



SAFETY FIRST! ALWAYS REMEMBER:

- Always ask a teacher or adult family member for help before starting any experiment!
- Use **small batteries** only—NEVER use wall outlets or plug into wall sockets!
- Never touch bare wires when they are connected to a power source.



Quick Recap - What We Learned Today

- Electricity flows like **water through pipes**.
- A **Voltage Divider** splits electricity to change brightness or power.
- A **Potentiometer** is a controllable knob used as a manual dimmer!



Student Practice Worksheet



Grade 4 Science - Voltage Divider & Potentiometer

Name: _____

Date: _____

Question 1: What does electricity flow like in our water pipe analogy?

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Question 2: What is the main purpose of a battery in an electrical circuit?

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Question 3: What does a voltage divider do to electricity?

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Question 4: What is a potentiometer? (Hint: Think of a special control item!)

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Question 5: What happens to a light bulb when you turn the potentiometer knob to let MORE electricity flow?

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Question 6: What happens to the light bulb when LESS electricity flows through it?

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Question 7: Name one real-life appliance in your house that uses a control knob like a potentiometer.

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Question 8: True or False: Resistors connected one after another are connected "in series".

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Question 9: Why do we need wires in a circuit?

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Question 10: What should you ALWAYS do before starting an electricity experiment?

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Question 11: Why is it dangerous to plug classroom or home science projects directly into a wall power outlet?

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Question 12: Fill in the blanks: Turning the potentiometer knob clockwise makes the light _____, while turning it counter-clockwise makes the light _____.

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